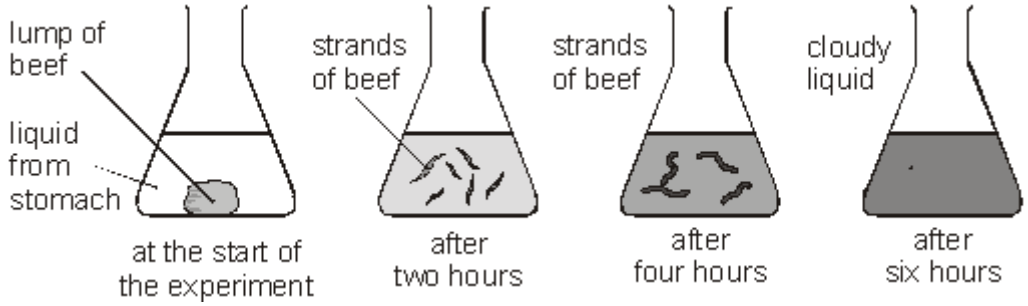
Q1.

A medical researcher used a tube to remove some of the liquid from a person's stomach. The researcher put the liquid into a flask which had a piece of boiled beef in it. The beef was digested in six hours. The diagrams show the flask at different stages of the experiment.



(a) (i) What type of substance in the liquid from the stomach caused the beef to be digested in six hours?

.....

1 mark

(ii) The researcher kept the flask at 37°C. Explain why.

.....

.....

1 mark

(b) In the human body, the digestion of protein in meat begins in the stomach and is completed in the small intestine.

(i) What is digested protein used for in the body?

.....

1 mark

(ii) Describe how the products of digestion get from inside the small intestine to the cells in the rest of the body.

.....

.....

.....

.....

2 marks
Maximum 5 marks

Q1.

(a) (i) an enzyme

*accept 'enzymes' **or** 'pepsin' **or** 'protease'*
*accept 'acid' **or** 'hydrochloric acid'*

1 (L7)

(ii) any **one** from

- it is the temperature at which enzymes work best
- it is the temperature of the body

*do **not** accept 'this is the body heat' **or** 'to make a fair test'*

1 (L7)

(b) (i) any **one** from

- for growth

accept 'for new cells'

- for repair
- for making other protein **or** enzymes

accept 'to supply amino acids'
*do **not** accept 'for energy'*

1 (L6)

(ii) **one mark is for the products of digestion getting through the barrier between intestine and blood, and the other mark is for the method of transport**

they pass through the intestine wall

*accept 'they are absorbed **or** pass into the blood'*
***or** 'they pass through the gut wall'*

1 (L7)

they are carried by the blood **or** plasma **or** blood stream

accept 'they go through the blood vessels
***or** arteries **or** veins'*

1 (L7)

[5]